



OPERATIONS MANAGEMENT

Final Report

Submission By:

Naimal Imtiaz

Junaid Khalid

Umar Alvi

Wasiq Ahmed

Table of Contents

<i>Submission By:</i>	0
1.1. Introduction	2
1.2. Objective	2
1.3. Mission	3
1.4. Vision	3
1.5. Core Values	3
1.6. Industry Overview	4
1.7. Key Products	5
1.8. Growth Factors	6
1.9. Industry Positioning	6
1.10. Organizational Structure	6
1.11. Markets Served	7
1.12. Data Collection Methods	7
1.13. Analytical Tools Used	7
1.14. Limitations of the Study	7
2.0. Processes	8
2.1. The Manufacturing Process	8
2.2. Quality Control	13
2.3. Technical Highlights	13
3.1. Problems & Solutions	14
4.1. Application of Operations Management Concepts	32
4.1. Conclusion	33
BIBLIOGRAPHY	34

1.1. Introduction

Newtech Industries (Private) Limited is one of Pakistan's major pipe manufacturers and has grown quickly within the plastic pipes market. The company started in 2000 and has expanded in response to the rising demand for good-quality galvanized and polyethylene pipes, especially HDPE, along with their fittings. Today, Newtech is widely known for dependable products and steady innovation. Its range includes HDPE, UPP, PVC, MDPE, PPRC, and electric conduit (EC) pipes, giving it a strong presence in the pressure pipes segment.

Quality has always been a central focus for the company. Newtech follows ISO 9001:2015 quality management standards, and its production processes are regularly tested in accredited laboratories. It is also the only pipe manufacturer in Pakistan that operates a full-time, round-the-clock R&D department, which helps the company maintain consistent quality and develop improved solutions over time.

Newtech's work can be seen in major projects such as Bahria Town, Green Gulberg, DHA Islamabad, and UNICEF-supported developments, reflecting its capacity to meet the needs of large infrastructure projects. With modern technology, updated production methods, and a strong customer-first mindset, Newtech continues to play an important role in shaping the future of Pakistan's pipe manufacturing industry, providing long-lasting and reliable solutions for the years ahead.

1.2. Objective

The goal of the visit to Newtech Industries is to understand the business, identify potential problem areas, and suggest practical improvements to enhance efficiency and production. The aim of understanding the company's processes is to identify operational problems and provide suggestions for optimizing performance at a high level.



1.3. Mission

Newtech Industries is dedicated to the pursuit of excellence, providing pipes and fittings that cater to the diverse needs of its customers. The objective of the company is both to modernize Pakistan's plastic pipe industry and to introduce new concepts while maintaining a customer-oriented vision. With a passion for excellence, Newtech aims to play its due role in the country's infrastructure and construction sectors, offering reliable, long-lasting, and sustainable products.

1.4. Vision

Newtech aims to become the frontrunner of the plastic pipe industry in Pakistan. Through the ongoing incorporation of customer feedback, adherence to international quality standards, and a culture of excellence, the company strives to achieve this goal. Guided by the motto "Committed to Excellence," Newtech is working to redefine industry benchmarks and emerge as an innovator in sustainable, high-quality solutions.

1.5. Core Values

- **Honesty & Ethics:** To conduct all business with integrity, in accordance with relevant law and regulations.

- **Uncompromising Quality:** Prioritizing quality to ensure conformance with international and domestic standards.
- **Exceeding Expectations:** Striving to surpass stakeholder expectations with innovative and reliable products.
- **Respect & Self-Esteem:** Valuing the contributions and dignity of all stakeholders, including employees, customers, suppliers, and shareholders.

1.6. Industry Overview

The pipe and fittings industry of Pakistan has experienced tremendous development in recent 20 years, which is due to the expansion of infrastructure, urbanization, and industrialization. There has been a massive demand increase for robust good quality pipe solutions in particular in field like water supply, sanitation, gas distribution and electrical conduit. As one of the key contributors to this emerging market, Newtech Industries (Private) Limited has been recognized as a pioneer of innovation, quality and customer satisfaction by consistently breaking records.

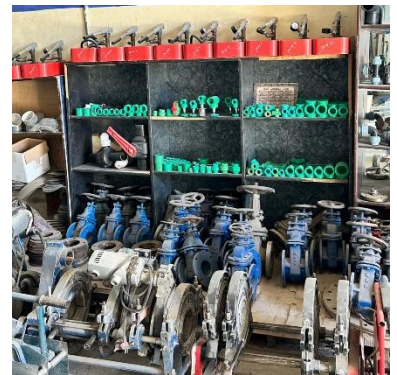


Newtech Industries is a large-volume firm in the pipe-making industry with wide-ranging manufacturing capacity and strong market dominance. The organization has more than 114 employees with a history of more than 320 projects. With 42 clients across the country, Newtech Industries has earned a reputation for offering high-quality, client-specific products and services.

1.7. Key Products

The industry is characterized by diverse product categories, including:

1. **High-Density Polyethylene (HDPE) Pipes:** Because of their robustness, ductility, and ability to withstand harsh environments, HDPE pipes are widely employed for water distribution, gas pipelines, and industrial applications.
2. **PVC pipes:** Lightweight, low-priced, and resistant to corrosion, PVC pipes are thus, preferred for plumbing and drainage applications.
3. **Medium-Density Polyethylene (MDPE) Pipes:** Basically, these pipes combine strength and flexibility. These pipes are suitable for gas distribution and potable water systems.
4. **Polypropylene Random Copolymer (PPRC) Pipes:** PPRC pipes have earned a reputation as a reliable performer at both high and low temperatures and are used for water supply in hot and cold-water systems.
5. **Electric Conduit (EC) Pipes:** These are non-metallic solutions that provide insulation and protective jackets for electrical cables and conductors.



1.8. Growth Factors

Key factors contributing to the expansion of this industry include:

- **Government Initiatives:** CPEC infrastructure development, new housing schemes, and urbanization projects significantly drive the demand for quality pipe systems.
- **Technological Advancements:** Adoption of standards such as ISO 4427 and ISO 9001:2015 ensures product quality and instills confidence among customers.
- **Sustainability:** A greater focus on eco-friendly and recyclable materials aligns with global sustainability efforts.
- **Customer Awareness:** Increased awareness among customers about the benefits of quality pipe systems in ensuring safety and longevity boosts the demand for high-quality products.

1.9. Industry Positioning

Newtech Industries is distinguished in the crowded market because of the following:

- **ISO Certification:** Compliance with international standards demonstrates the company's commitment to quality.
- **Innovative Solutions:** With a dedicated R&D team, Newtech launches state-of-the-art products that cater to current construction and industrial needs.
- **Strong Partnerships:** Collaborations with leading contractors and projects reflect the company's dependability and competence.

1.10. Organizational Structure

The hierarchical structure under which the company operates includes the CEO, Operations Manager, Production Supervisors, R&D Team, Quality Assurance, and Workforce.

1.11. Markets Served

Newtech Industries (Private) Limited serves a wide range of markets, with a focus on the construction, infrastructure, and utilities sectors. The company's high-quality pipes and fittings are used in various landmark projects across Pakistan, including residential and commercial developments such as Bahria Town, DHA Islamabad, and Green Gulberg. Additionally, Newtech provides robust piping solutions for public works projects executed by the National Highway Authority (NHA) and Frontier Works Organization (FWO). With a portfolio that includes international organizations such as UNICEF, Newtech ensures its products meet the exacting demands of both local and global markets, delivering reliability and excellence in every application.

1.12. Data Collection Methods

- Interviews with operations manager and workforce
- Site visits to manufacturing facility (*Images showing on site visit in Appendix*).
- Observation of production and storage processes
- Review of internal documents and inventory logs

1.13. Analytical Tools Used

- Flowcharts for process mapping
- Fishbone (Ishikawa) Diagrams for root-cause analysis
- 5 Whys Technique
- Demand forecasting with Exponential Smoothing
- Six Sigma & Total Quality Management (TQM) principles
- Just-in-Time (JIT) production analysis

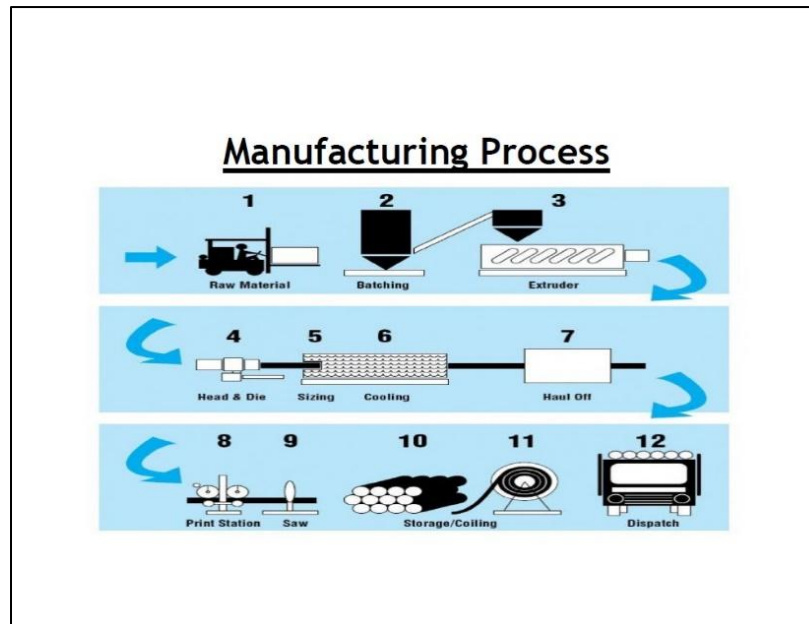
1.14. Limitations of the Study

- Access restricted to certain confidential documents
- Limited observation period may not cover all seasonal fluctuations

- Data dependent on employee reporting accuracy

2.0. Processes

2.1. The Manufacturing Process



2.1.1. Raw Material

The process starts with the selection and preparation of raw materials. For pipe manufacturing, common materials include:

- **PVC (Polyvinyl Chloride):** Hard and strong, used in plumbing and drainage systems.
- **HDPE (High-Density Polyethylene):** Flexible, lightweight, and resistant to chemicals, suitable for gas and water pipelines.
- **PP (Polypropylene):** Heat-resistant and used for industrial piping.

The unprocessed inputs, in granular or fine powders, are stored in silos or bins with environmental control to ensure the inputs are not contaminated, and the integrity of the

inputs is preserved. Quality control tests, e.g., melt flow index (MFI) determinations, guarantee that the material has the required properties prior to use.

2.1.2. Batching

Batching involves careful titration and tarrying of raw materials with mix mates. The additives used in pipe manufacturing can include:

- **Stabilizers:** Improve heat resistance and durability.
- **Plasticizers:** Increase flexibility (used in non-rigid pipes).
- **Pigments:** Provide color and UV resistance.
- **Lubricants:** Reduce friction during extrusion.

Automated batching systems weigh and blend these materials based on formulations designed for particular pipe uses. The blend is sonicated to produce consistent additive distribution, which is a key aspect for achieving consistent pipe properties.

2.1.3. Extruder

Extruder is the core machine of the pipe manufacturing line. The raw material blend is added to the extruder's hopper, where the following takes place:

- **Melting:** In the extruder a rotating screw drives the material through heated passages. Temperatures are carefully controlled, typically ranging from 160°C to 230°C depending on the material.
- **Compression:** The material is compressed to remove air voids, which leads to uniform melt.
- **Mixing:** Shear stresses resulting from the screw homogenization entraps the molten material.

Current extruders can be equipped with a single or twin screws and with twin-screw extruders typically providing superior mixing and throughputs for complex

formulations. The molten plastic escapes from the extruder under high pressure into the die.

2.1.4. Head & Die

Molten plastic flows into an extrusion die, a highly specialized tool which allows the plastic to be formed into a continuous tube shape. The key components here include:

- **Die Design:** Determines the pipe's diameter and wall thickness. Spiral mandrel die of PVC pipes is frequently employed to guarantee uniform flow distribution.
- **Calibration Sleeves:** Ensure that the pipe's dimensions match specified tolerances.

Die temperature and pressure are both important to reduce defects such as warping or wall thickness inhomogeneity. Sophisticated dies can be equipped with features for processing multilayer/co-extruded pipes.

2.1.5. Sizing

Sizing involves setting the final dimensions of the pipe. This step typically occurs in a vacuum calibration tank. Here's how it works:

- **Vacuum Chamber:** The tube is located in such a way that a negative pressure inside the chamber keeps the tube in a constant configuration.
- **Water Spray or Baths:** Help cool the pipe and lock in its dimensions.
- **Calibration Sleeves:** Maintain uniform wall thickness and circularity.

Vacuum sizing is a critical factor for achieving tight dimensional tolerances, which dictate pressure resistance or flow rate requirements, depending on the application.

2.1.6. Cooling

Following size, the pipe is cooled to set its configuration. Cooling can be done using:

- **Water Baths:** Submerging the pipe in water for rapid cooling.
- **Spray Cooling:** Jets of water sprayed over the pipe's surface.

Cooling curves are optimized in order not to induce internal strains that may impair the integrity of the pipe. For large-bore pipes, multi-stage cooling systems are employed in order to achieve a uniform temperature decrease.

2.1.7. Haul Off

The belt-off unit allows the pipe to keep moving in the production line. Key components include:

- **Traction Belts or Rollers:** Grip the pipe without damaging its surface.
- **Variable Speed Control:** Compares extruding speed in order not to overstretch or over compress the pipe.

Haul-offing machines play a very important role in guaranteeing production reproducibility, especially for the pipes which are made with long lengths or thick walls.

2.1.8. Print Station

At the print station, relevant information is inscribed on the surface of the pipe by means of inkjet or laser printers. Common information includes:

- **Manufacturer name.**
- **Pipe diameter and thickness.**
- **Material type** (e.g., PVC-U, HDPE).
- **Batch number and certification marks.**

Ink used in printing must be weatherproof and durable to withstand handling, transport, and prolonged exposure to the environment.

2.1.9. Saw

The pipe is of defined lengths with a cutting mechanism (which may be) [1].

- **Rotary Saw:** Ideal for smaller diameter pipes.

- **Planetary Cutter:** Used for large-diameter or thick-walled pipes.
- **Guillotine Cutter:** Provides clean, precise cuts.

Sophisticated cutters can extend to automating cut lengths adaptively with production needs.

2.1.10. Storage/Coiling

Once cut, pipes are either:

- **Bundled:** For rigid pipes, they are stacked in uniform bundles.
- **Coiled:** Flexible pipes (e.g., HDPE) are wound into coils for more efficient transportation and installation.

Special care is taken during storage to prevent deformation, particularly for thermoplastics susceptible to creep under prolonged loading.

2.1.11. Storage (Continued)

Stored pipes are shielded from environmental exposures such as sun (which can cause damage to UV-activated material) and moisture. Pipes are usually wrapped with protective sheeting or stored under a cover when they are to be stored for a long time.

2.1.12. Dispatch

The final step involves loading the pipes for transportation. Logistics crews see that the pipes are properly shipped by securing them to prevent damage. Some key considerations during dispatch include:

- **Weight Distribution:** Proper stacking to avoid crushing pipes at the bottom.
- **Labeling:** Ensures that batches are correctly identified and traceable.
- **Compliance with Regulations:** Pipes are inspected for quality and certification before shipping.



2.2. Quality Control

Throughout the process, strict checks for quality are performed to guarantee pipes comply with industry standards such as ASTM, ISO, or DIN. Key tests include:

- **Dimensional Accuracy:** Ensures consistency in diameter and wall thickness.
- **Hydrostatic Pressure Testing:** Verifies the pipe's ability to withstand internal pressure.
- **Impact Resistance:** Assesses the pipe's durability under mechanical stress.
- **Material Testing:** Confirms the chemical composition and properties of raw material

2.3. Technical Highlights

1. **Automation:** Automation is increasingly being applied to almost every step of the modern pipe manufacturing process, from batched mixing to cut settings, to enhance efficiency and decrease the liability of human error.
2. **Energy Efficiency:** Cooling and extrusion system designs are intended to be energy efficient while producing high quality goods.

3. **Customization:** Dies and calibration systems can be adjusted to produce pipes for various applications, including water supply, drainage, or industrial use.

This detailed process flow ensures that the pipes produced meet the highest standards for safety, durability, and performance.

3.1. Problems & Solutions

3.1.1. Overstocking of Pipe Fittings and Machinery Underutilization at Newtech Industries

Overstocking of pipe fittings in storage at Newtech Industries is one of the major operational issues recognized. The company currently has an oversupply of some pipe fittings which in turn means the machines for manufacturing these fittings are not fully utilized. Current situation reveals inefficiencies of production planning, inventory management and demand forecasting procedures.

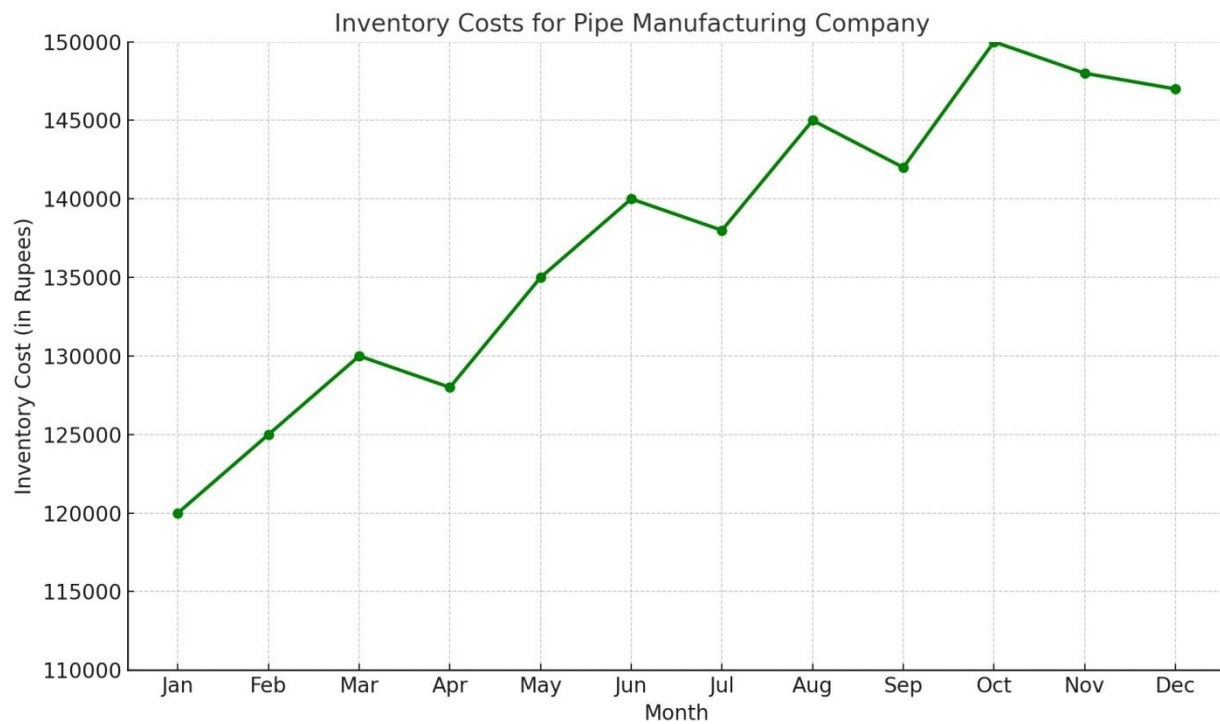
Identification of the Problem

This problem was identified during our factory visit and subsequent interactions with the operations manager and workers. Observations revealed idle machinery in the production area, and feedback from stakeholders highlighted the mismatch between production planning and market demand. Additionally, a search of inventory logs verified the problem of overstocking. The use of imported specialized pipe fittings was shown to be an effective plan decision based on the high cost of getting the associated equipment.

3.1.1.a. The Issue of Overstocking

Overstocking occurs when the quantity of products stored exceeds the actual market demand. In the case of Newtech Industries, the surplus stock of pipe fittings is a direct result of inadequate demand forecasting and production planning. Excess inventory ties up financial resources, increases storage costs, and risks product obsolescence.

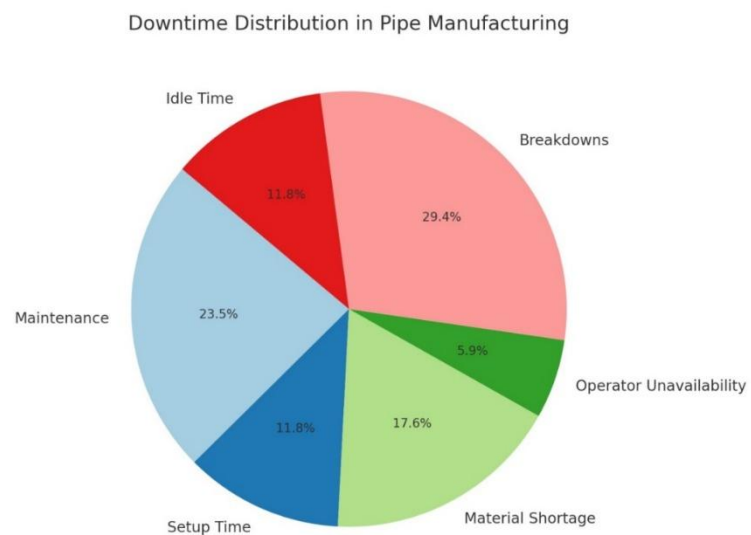
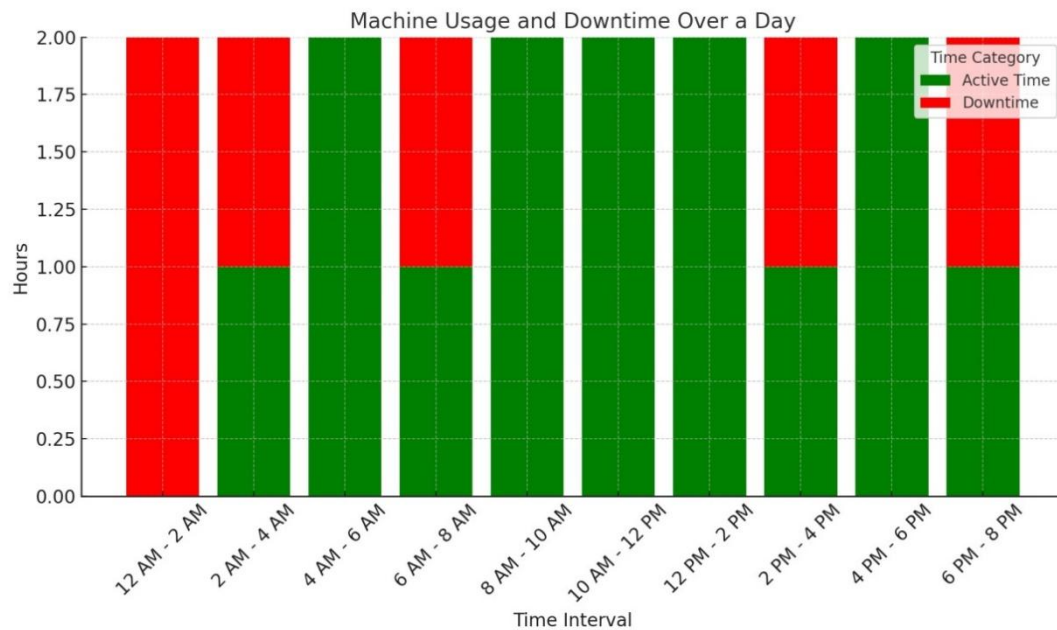
Additionally, the storage space consumed by overstocked items reduces the company's capacity to handle other essential materials or products, creating further logistical challenges.



Month	Inventory cost in PKR
January	120,000
February	125,000
March	130,000
April	128,000
May	135,000
June	140,000
July	138,000
August	145,000
September	142,000
October	150,000
November	148,000
December	147,000

3.1.1.b. Underutilization of Machinery

The machinery used for manufacturing these overstocked pipe fittings is often idle because of the existing surplus. This underutilization leads to a waste of resources and capital, as expensive equipment is not being used to its full potential. Machinery that is not in operation still incurs costs in terms of depreciation, maintenance, and opportunity loss, which compounds the financial burden.



3.1.1.c. Dependency on Imports

Adding to the complexity, Newtech Industries imports a substantial portion of its pipe fittings because it lacks the machinery required to manufacture certain designs. While importing may be cost-effective compared to investing in high-cost machinery, it also introduces supply chain vulnerabilities such as longer lead times, fluctuating exchange rates, and dependency on external suppliers. This dependency limits the company's ability to respond swiftly to changes in customer demand and creates inefficiencies in the overall supply chain.



Root Cause Analysis

The underlying reason for these problems can be explained using the perspective of the Theory of Constraints (TOC). TOC defines the bottleneck or restriction of a system that confines the overall performance of a system. In this case, the bottleneck lies in two areas:

- 1. Deletion of accurate demand forecasting and resulting overproduction and overstocking.**
- 2. Lack of appropriate machine to manufacture demanding pipe fitting designs domestically, in turn prompting reliance on imports.**

The problem of overstocking is exacerbated by a lack of integration between production and market demand. By means of TOC, it is evident that what constrains is not the capability of production, but the discrepancy between the planned production and the real demands of customers. The malfunctioning of such existing equipment is a manifestation of this misalignment.

Solving Overstocking and Machinery Underutilization Using Just-In-Time and Exponential Smoothing Principles

Operational problems associated with overstocking and machine idleness at Newtech Industries can be readily solved by the “Just in Time” (JIT) principle and Exponential Smoothing. These approaches stem from operations management theory and deliver evidence-based, data-driven, structurally sound, and actionable solutions to enhance efficiency and match production to demand.

Just-in-Time (JIT) Principle

The Just-in-Time principle operates on the basis of reducing inventory by manufacturing and buying goods when they are required for immediate use. Implementing JIT at Newtech Industries would require reorganization of inventory control practices and production flows to minimize warehousing and allow materials and end products to flow through the production system as smoothly as possible.

Steps to Implement JIT:

1. Streamline Production Scheduling:

Production should be matched to real world customer orders or true demand forecasts. This ensures that pipe fittings are manufactured only when there is a confirmed need, thereby preventing overstocking.

2. Reduce Lead Times:

Close collaboration with suppliers can shorten the time it takes to acquire raw materials. To import pipe fittings, this may require the establishment of contracts with trustworthy sources to guarantee that the products arrive on time.

3. Set Inventory Thresholds:

Implement a system that triggers production or procurement when inventory levels reach a specific threshold. This prevents unnecessary buildup of stock while ensuring there is no shortage.

4. Enhance Worker Accountability:

Each stage of production should be carefully screened for the possibility of bottleneck and delay. Training programs may assist workers with the knowledge of how their work can influence the effectiveness of JIT operations.

Expected Outcome:

Through the use of JIT, Newtech Industries would develop a leaner inventory, thereby decreasing inventory carrying costs and eliminating the waste characteristic of holding obsolete material in excessive inventory levels. Additionally, this system would increase machinery utilization by producing items only in response to demand rather than speculative production.

Exponential Smoothing and Its Application as a Solution

Exponential Smoothing is a statistical technique used for demand forecasting that assigns exponentially decreasing weights to past observations. This implies the recent data points are assigned greater weights than the historical ones, in favor of increasing the responsiveness of the forecast to demand pattern changes. In contrast to simple moving averages, Exponential Smoothing can take into account the informativeness of recent trends not necessarily making the analysis unstable by averaging out of short-term variations.

Exponential Smoothing can be used in the framework of Newtech Industries in order to solve overstocking and underutilization of machinery problems. The problem of overstocking is caused by lack of good demand forecasting, so pipe fitting production could be overloaded. Using Exponential Smoothing, the company can sum up not only the historical sales information but also infer it as an estimation of the demand, while at the same time it can respond to the recent market trend. This prevents overproduction risk and matches production schedules to demand as it actually exists.

In addition, reliable demand forecasting guarantees optimal machine utilization. Exponential Smoothing, rather than manufacturing pipe fittings "in anticipation" of demand, which may not exist, allows data-driven decision making. This means that the company can, for example, tailor its production cycles, so that machine operation is limited to times when it is really needed to respond to demand forecasted orders. Consequently, Newtech Industries is not left with idle (machine) time and also decreases storage costs by minimizing (excess) inventory, thus, resulting in a more balanced (and efficient) production setup.



3.1.2. Lack of Worker Motivation and Its Impact on Operations

An important problem in Newtech Industries is that of low worker motivation. According to the operations manager, the workers' tasks have been reduced to relatively simple operations due to the highly automated nature of the production process. The main functions of the workers are machine controlling, allocation as pipe length (and width), and setting as laser engraving etc. Yet, workers sometimes fail to recognize the major role that they play. This gap in knowledge, intermixed with the repetitive nature of their jobs, has resulted in a decrease of motivation and work engagement.

Understanding the Problem

Workers' disinterest stems from a disconnect between the perceived value of their tasks and the actual consequences of errors. Although their responsibilities may appear straightforward, even minor mistakes—such as inputting incorrect dimensions or engraving settings—can result in significant financial and operational repercussions. For example, an incorrect calibration in pipe dimensioning might make an entire batch unusable, because it leads to wasted material, extra costs for reworking, and delays in satisfying customer orders.

This problem is most acutely felt when demand is high and there is little room for error. Inadequate focus on detail of the operatives or undervaluation of the contribution of the operative to the production process can result in production bottlenecks, in turn hindering ability of the company to provide on time. Although their contributions are of utmost importance, workers tend to be blind to the bigger operational picture, which in turn reduces their feeling of responsibility and involvement.

Applying Herzberg's Motivation-Hygiene Theory

Herzberg's Motivation-Hygiene Theory provides a robust framework for understanding and addressing the lack of worker motivation at Newtech Industries. According to the theory, two sets of factors influence employee satisfaction and motivation: **hygiene factors** and **motivators**.

1. **Hygiene Factors:** These are extrinsic elements related to the work environment, such as salary, working conditions, company policies, and job security. Although these factors by no means lead to motivation, their absence can result in a sense of dissatisfaction. However, at Newtech Industries, the workers' dissatisfaction may not be caused by insufficient hygiene factors since their work environment and the degree of job security seem adequate. Nevertheless, the lack of intrinsic drive may account for their motivation decline.
2. **Motivators:** Intrinsic aspects of the type of work, such as, achievement, praise, duty, and opportunities to develop and learn, are here. Motivators are critical for fostering job satisfaction and engagement. Here, workers do not have chances for accomplishment and reward, because their jobs are considered mundane and of no value. Further, the lack of a clear idea about the effect of their roles depletes the feeling of accountability. Intervention and the application of the total quality management (TQM) Principle to worker motivation.

Addressing Worker Motivation Using the Total Quality Management (TQM) Principle

Total Quality Management (TQM) represents an integrative management model for integrating the culture of quality into every organizational process. The concept highlights the importance of continuous feedback, customer satisfaction and engagement of the employees in attaining quality goals. TQM is based on a structured attention to teamwork, process improvements, and a shared goals and burden of responsibility in all aspects of the organization.

Quality forming the heart of TQM is assumed to be not the responsibility of certain department but the responsibility of the whole organization. All employees, no matter what they do, are involved in the quality of goods and services. Through the focus on training, teamwork, and empowerment, TQM aims to map the workforce to the organization's quality objectives while fostering employees' pride in their work and their ability to realize its contribution to success.

Application of TQM to Address Worker Motivation

In the case of Newtech Industries, the introduction of TQM principles is able to effectively solve the problem of the lack of motivation of the workers. Although by the automated production line, workers' physical strain is reduced, the necessity of personnel to maintain quality control is not less. Missteps in entering machine settings or overseeing production can result in defects, wasted resources, and delayed timelines. TQM offers a framework to reframe workers' roles, emphasizing their importance in the larger operational and quality landscape.

1. Enhancing Employee Involvement

Emphasis in TQM is that employees are a part of quality maintenance and enhancement. Through involvement of work in the process of quality control decision making, Newtech Industries can generate ownership and responsibility. For instance, workers could participate in regular quality review meetings where they share feedback on machine operations, discuss challenges, and propose solutions. That engagement not only improves transactions but also reciprocates and affirms workers, providing a motivational satisfier.

2. Training and Skill Development

A core tenet of TQM is continuous learning. Providing organized training programmes can enable workers to realize the importance of their work and how their work fits into company quality targets. At Newtech Industries, training sessions can highlight the potential consequences of errors in machine settings and demonstrate best practices for maintaining precision. Training further enables the worker to develop feeling of competency which is the main propellant for intrinsic motivation.

3. Implementing a Recognition System

The practice of recognizing and motivating employees' quality contribution is also of great importance TQM. Worker, who always show an eye for the details and maintains

the quality standard, should be rewarded, certificates or name and fame. By celebrating successes, Newtech Industries can create a culture of appreciation, encouraging workers to remain engaged and diligent in their tasks.

4. Embedding a Continuous Improvement Culture

TQM promotes the idea of continuous improvement, where every employee is encouraged to identify inefficiencies and suggest improvements. Whenever there are workers operating machines, this can include making a report on repeated problems, suggesting improvements to how the machine is operating, or trying out done 'work' to improve a process. If their proposals are realized, it increases their feeling of utility and drives their desire to make additional contributions.

5. Promoting Quality Awareness

In order to offset the mismatch between employees and their world, Newtech Industries must stress that quality must always be at the heart of the production process. Such as quality-awareness training sessions, visual cues in factories (e.g., posters displaying scope and cost of cost of errors) and dashboards that present real-time quality statistics). If the employees' tasks are correlated to the success of the company, they will be more apt to understand their contribution to success, thereby developing a deeper feeling of responsibility.



3.1.3. Salt Lines on Pipes Due to Manual pH Adjustments

Another substantial problem at Newtech Industries is the generation of salt tracks in pipes during production. This issue occurs when the pH of the cooling tanks is not properly balanced. Cooling is an important process for guaranteeing the longevity and quality of pipes, and any variation in the desired water pH can cause the formation of salt deposits due to the presence of ions in the water on the pipes. These deposits make the pipes unusable, causing financial loss and production blockage.

The Newtech Industries cooling method consists of running pipes through water tanks in order to cool down the pipes after they are produced. This step is very important for the integrity of the structure and the quality of the pipes. But the water acidity level has to be kept within a certain range to prevent salt build ups in the pipe wall.

Currently, the company relies on manual pH adjustments. Measurements of pH level are made periodically by the workers and chemicals are added when necessary to ensure balance. This approach introduces several inefficiencies and risks. Accident of pH measurement or pH adjustment is one of the most frequent phenomena, especially when pH measurement or pH adjustment is a recurrent operation. In addition, abnormal intervals between pH monitoring may permit pH value deviations to be missed for longer durations.

Salt streaks on pipes resulted as a direct consequence of such failures. Salt lines not only spoil the look of the pipes, but also cause customers to worry about the durability of the pipes. A faulty batch because of salt deposit in its pipes can cause great economic losses, such as material loss, high replacement production costs, and damage to the company's market reputation.

In addition, the absence of automation in such a process is a lost opportunity for process optimization. With current quality control systems, it is possible to remove the risks of manual pH adjustment by means of real-time monitoring and automatic corrections. The

lack of such mechanisms highlights the urgent requirement for sustained investment in next generation technology, in order to protect the quality of production.



Fishbone Diagram Analysis

The **Fishbone Diagram** below highlights the root causes of the issue by categorizing factors contributing to salt line formation:

1. Materials:

- Water used in the cooling tanks may have inconsistent pH levels, requiring regular monitoring and adjustment.
- No automated system exists to ensure consistent water quality.

2. Methods:

- The pH adjustment process is entirely manual, increasing the likelihood of human error.
- Inconsistent testing intervals fail to address fluctuations in pH levels.

3. Machines:

- The cooling tanks lack integrated systems for automated pH monitoring and adjustment.
- Machines do not provide real-time alerts for pH deviations.

4. People:

- Workers may not have sufficient training to understand the consequences of improper pH adjustment.
- Dependence on manual checks increases the risk of oversight or error.

5. Environment:

- Production floor conditions, such as time constraints and workload, can distract workers from performing precise measurements.

5 Whys Technique

Problem: Salt lines are forming on pipes after passing through the cooling tanks.

1. Why are salt lines forming on pipes?

Because the water pH levels in the cooling tanks are not consistently maintained.

2. Why are water pH levels not consistently maintained?

pH levels are manually adjusted, creating a possibility for human error.

3. Why are pH levels adjusted manually?

Due to the lack of an automated system to track and control real-time pH levels.

4. Why has an automated system not been implemented?

Due to the company's history of focusing cost-cutting techniques over adopting state-of-the-art quality control technology.

5. Why is cost-saving prioritized over quality control?

Due to the dominance of short-term financial targets over the optimization of long-term processes.

Implementation of Six Sigma

Six Sigma can be implemented to enhance the quality and consistency of pipe production by systematically reducing defects and process variations. For instance, the recurring issue of salt lines forming on pipes due to manual pH adjustments in cooling tanks can be addressed through the Six Sigma DMAIC (Define, Measure, Analyze, Improve, Control) methodology. The problem is clearly defined as defects in finished pipes, and data is collected on pH levels, frequency of manual adjustments, and incidence of salt lines. Analysis identifies root causes, such as inconsistent monitoring intervals and human error. Improvements can be made by automating pH measurement and adjustment, integrating real-time sensors, and standardizing procedures. Finally, control measures such as continuous monitoring dashboards and periodic audits ensure the process remains stable, reducing defect rates and enhancing overall product quality, while aligning production efficiency with international best practices.

Applying Total Quality Management to the Problem

1. Focus on Process Improvement:

- TQM defends the need for process optimization to pursue continuous quality. Automatic pH control in cooling tanks is in line with this objective, as it avoids variability and human error in an important quality control process.

2. Implementing Quality Control Systems:

- TQM promotes the application of technology to improve quality assurance. Installing automated pH monitoring and dosing systems ensures that water quality is maintained within the desired range at all times, preventing salt line formation on pipes.

3. Employee Training and Awareness:

- Workers play a vital role in TQM. Training employees to operate the automated system and to appreciate the significance of maintaining pH levels in water encourages an ownership of quality assurance roles. This also puts their jobs in support of the company's more general quality commitment.

4. Continuous Monitoring and Feedback:

- In TQM, ongoing surveillance and data-based decisions are important. Real-time carbonate measuring feedback from auto sampling features makes the information easily interpretable in order to detect and attend to problems early on.

5. Preventive Maintenance:

- TQM promotes proactive measures rather than reactive fixes. Regular maintenance schedules for the automated systems ensure their reliability and longevity, reducing downtime and preventing production disruptions.

3.1.4. Inadequate Storage Facilities for Pipes

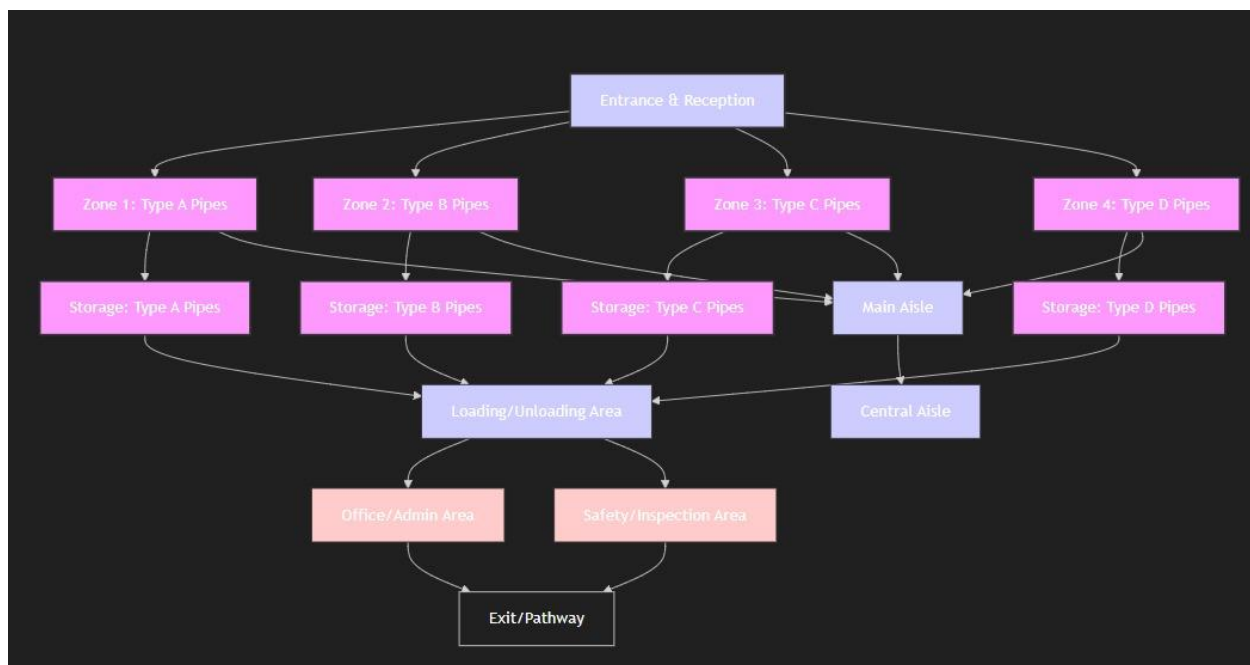
One of the operational inefficiencies identified at Newtech Industries is the lack of proper storage facilities for finished pipes. Post production, pipes are moved to a warehouse that is not properly equipped with infrastructure (e.g., roof) to weatherize the products to external environmental factors. This, consequently, puts the pipes in contact with dust, sunshine and other external aggressors which can damage them and reduce their quality and longevity. In addition to the above, the pipes are piled and bound together in an uncontrolled fashion, rendering them an even greater potential target of damage during picking up and stowing.

Pipes exposed to sunlight may experience material degradation over time, particularly if they are made from materials sensitive to UV rays. Similarly, exposure to dust not only impacts the pipes' appearance but can also require additional cleaning before delivery to

customers, increasing operational costs. Additionally, stacking and tying pipes without a structured storage system increases the risk of physical damage. Pipes can be bent, scratched, or otherwise deformed by the mishandling, resulting in material loss and may cause delays in the delivery of customer orders. Not only these procedures have an impact on the quality of the finished goods but these sensational practices also threaten the company's image for supplying quality goods.

The lack of adequate storage infrastructure also plays a role in inefficiencies for inventory management. If there is no systemically defined and structured system, that is, it is unstructured then it is hard to gain an accurate inventory of inventory levels, which may lead to stockouts or excess stock. This mismanagement may lead to shipment deferments, customer displeasure and, moreover, higher operational costs.

Facility Layout Design



Inadequate storage auto sampling problem can be studied and solved by way of **Facility Layout Design**. Facility Layout Design deals with the best physical restructuring of resources, processes, and space of a given organization to achieve maximum efficiency

and optimized performance of the system. One of the core tenets of this principle is a supporting storage infrastructure, because it directly influences inventory control, product quality, and the efficiency of the operations.

1. Flow of Materials:

The absence of an organized storage facility disorganizes the smooth supply of items between production and receipt. Exposed pipes provide lines of blockage in the supply chain, because further processing is often required for cleaning, rectification, or replacement of damaged items.

2. Space Utilization:

Current storage systems do not maximize available space. Pipes are haphazardly piled and there is no space for an inventory to be properly tracked and retrieved. Inefficient use of space can result in longer handling time & labor cost.

3. Protection of Products:

Facility Plan Design stresses protection of the final products to prevent quality deterioration. The absence of a roof and protective storage solutions fails to shield pipes from environmental damage, compromising their usability and leading to potential financial losses.

4. Inventory Management:

Disorganized storage hampers accurate inventory tracking. Facility Layout Design integrates storage systems that allow for better categorization and identification of products, ensuring inventory levels match demand and reducing the risk of overstocking or shortages.

Proposed Solutions

1. Construct a Covered Storage Facility:

Construct a covered barn to shield pipes from sunlight and dirt. Include ventilation systems in order to provide airflow and keep environmental pollutants from entering.

2. Introduce Racks or Shelving Systems:

Provide vertical storage options (e.g., racks) for pipe storage in size, material, or production date order. This optimizes space utilization and minimizes the risk of damage during handling.

3. Designate Clear Storage Areas:

Chunk the storage space into discrete areas that are easily identifiable by product classes or order importance. Employ labeling systems and digital inventory instrumentation for proper identification and access.

4. Integrate Material Handling Equipment:

Move pipes about the warehouse using fork trucks or conveyor belts eliminating manual handling and the risk of damage which may be associated with it.

4.1. Application of Operations Management Concepts

OM Concept	Theoretical Best Practice	Current Practice	Formal/Informal	Gaps Identified
Process Design	Optimized, standardized workflows	Detailed extrusion, sizing, cooling, and finishing steps	Formal	Manual pH adjustment, storage inefficiencies
Capacity Planning	Match machine capacity with demand	Some machines underutilized	Informal	Idle machinery due to overstocking
Demand Forecasting	Data-driven methods	Largely manual or historical data	Informal	Overproduction of certain fittings

Inventory Management	JIT & automated tracking	Overstocked inventory, poor storage	Informal	High storage costs, obsolescence risk
Quality Management	TQM, Six Sigma, automated monitoring	Manual pH, limited automated QC	Partial	Salt lines on pipes, human error
Workforce Management	Engagement, training, recognition	Limited intrinsic motivation	Informal	Low morale, errors
Supply Chain & Logistics	Integrated, responsive	Dependent on imports	Partial	Delays, risk exposure

5.1. Conclusion

Guided by the fundamental principles of integrity, unyielding quality, being above and beyond the customer's expectation and treating everyone with respect, Newtech Industries has become a leading force in this constantly changing market, achieving new standards of innovation, quality, and customer satisfaction. While the company processes operational challenges such as stock excess in pipe fittings and underutilization of machine, implementation of Just-in-Time (JIT) principles and Exponential Smoothing it can be seen as an ordered, data-driven method to maximize efficiency and synchronize production with demand. In addition, using Total Quality Management (TQM) principles to motivate the workforce and to overcome problems, for example, the formation of salt deposits on pipes caused by manual pH control by means of process improvement and automation, Newtech Industries will be able to strengthen this company's position as a leader of sustainable and high-performance solutions, thus guaranteeing the company's solid and competitive position in the Pakistani plastic and pipe market.

BIBLIOGRAPHY

- Gangidi, P. (2019). A systematic approach to root cause analysis using 3× 5 why's technique. *International Journal of Lean Six Sigma*, 10(1), 295-310.
- Harrison, P. J. (1967). Exponential smoothing and short-term sales forecasting. *Management Science*, 13(11), 821-842
- Desai, K. J., Desai, M. S., & Ojode, L. (2015). Supply chain risk management framework: a fishbone analysis approach. *SAM Advanced Management Journal*, 80(3), 34
- Schonberger, R. J. (1982). Some observations on the advantages and implementation issues of just-in-time production systems. *Journal of Operations Management*, 3(1), 1-11
- Gillis, A. S., & Pratt, M. K. (2023, April 5). What is Total Quality Management? | Definition from TechTarget. Search CIO; TechTarget. <https://www.techtarget.com/searchcio/definition/Total-Quality-Management>